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MNT/004 – Policy for the Inspection and Maintenance of Overhead Systems

1 Purpose

The purpose of this document is to state the company's policy for the inspection and maintenance of overhead lines and pole mounted plant. Further to this, the policy describes key assumptions made in the preparation of this document and also sets out to provide clarity across the organisation with respect to alignment of practice across legacy companies.

This policy also helps to ensure the company achieves its requirements with respect to The Electricity Safety, Quality, and Continuity Regulations 2002, and the Electricity at Work Regulations 1989.

This document supersedes the documents detailed in Section 3.9, all copies of which shall be destroyed.

2 Scope

This document encompasses all overhead lines and pole mounted plant at LV, HV, and EHV and describes the general inspection and maintenance requirements for the asset group. The detailed inspection and maintenance tasks are not provided within the scope of this policy document.

The general asset groups covered within this policy are:

- overhead line conductors,
- overhead line supports; and
- pole mounted plant.

3 Policy

3.1 Assessment of Relevant Drivers

The key business drivers relating to the inspection and maintenance of overhead lines are:

- safety,
- financial; and
- quality of supply.

Further, the company has a duty to comply with the Electricity Safety, Quality and Continuity Regulations 2002 to:

- ensure our equipment is “...so constructed, installed, protected (both electrically and mechanically), used and maintained as to prevent danger, interference with or interruption of supply, so far is as reasonably practicable”; and
- “...inspect [our] network with sufficient frequency so that [we are] aware of what action we need to take so as to ensure compliance with these Regulations”

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3.2 Key Requirements

The objectives of this policy are to:

- ensure that overhead lines and pole mounted assets are kept in an acceptable condition for continued service on the network whilst minimising the impact on quality of supply to our customers and avoiding the financial impact of a failure that could be detected before it occurs; and
- safeguard members of staff, contractors and the general public from the hazards that are posed by the overhead line asset base.

3.3 EHV Overhead Lines

Here, 'EHV' covers the range from 33 to 132kV inclusive.

3.3.1 Summary of Intervals

Activity	Asset Type		
	EHV wood pole lines	EHV masts, towers and rolled steel supports	YEDL single girder 'Woodhouse' masts
High risk site inspection *	2 Years	2 Years	2 Years
Stage 1 condition assessment	2 Years	2 Years	2 Years
Thermal Imaging	4 Years	4 Years	4 Years
Stage 2 condition assessment	10 Years	10 Years	10 Years
Helicopter Patrol (High Resolution Imaging)		10 Years	none †

Table 3.3.1

Notes:

1. Inspections shall be combined as appropriate. For example, a high risk site shall receive both a stage 1 condition assessment and a high-risk site inspection every 24 months, and these shall be undertaken as a single visit wherever practicable. Similarly, every fifth inspection shall be a full stage 2 condition assessment
2. All condition and risk assessment data to be collected during inspections shall be entered onto an appropriate inspection sheet proforma or hand held device, then deprogrammed onto a database readily accessible across the business using standard asset identifiers

* Further details on the risk classification of overhead line sites are provided in the policy for the Risk Assessment of Overhead Lines (DSS/023). Amongst other things, this provides that all steel lattice supports shall be considered high risk

† Woodhouse steel masts do not require helicopter inspection with high resolution imaging since they shall be programmed for replacement with a modern equivalent overhead line design. Masts shall only receive a stage 1 and 2 condition assessments and, where appropriate, high risk site inspections in line with the intervals shown until they are eradicated from the distribution network

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3. Following any lightning storm, it is recommended that a localised visual inspection of surge arresters is carried out.

3.3.2 High Risk Site Inspection

High risk sites shall be inspected in accordance with the intervals shown in table 3.3.1 above. During these inspections, particular attention shall be paid to:

- evidence of trespass or loitering;
- evidence of vandalism;
- classification of the risk associated with equipment;
- classification of the risks associated with the surrounding land;
- safety related assessment of equipment condition; and
- assessment of the existing mitigating measures at that site.

The detailed requirements for the high risk site inspection are described in the policy for Risk Assessment of Overhead Lines (DSS/023).

A high risk site where unauthorised access, trespass, theft or vandalism is evident will be classified as vulnerable. More frequent inspections of these sites shall be undertaken to monitor their status. The frequency of visits shall be determined depending on the particular risk that is evident. Information from RCM studies indicate that a quarterly inspection would be appropriate for these sites, although more frequent inspections may be required in particular circumstances.

All sites with overhead line inadvertent contact warning signs shall receive an annual warning sign inspection as described in the policy for Risk Assessment of Overhead Lines (DSS/023). Such signs, and standard safety signs, shall be replaced if they are considered insufficient to last until the next inspection. All work shall be in accordance with NSP/004/109 "Guidance note on ACDs safety signs and labels required on overhead lines".

Consideration shall be given to removing the site from the vulnerable site register when no evidence of trespass activity is evident for more than two consecutive visits or additional control measures are employed.

3.3.3 Stage 1 Condition Assessment

Lines shall be patrolled by helicopter or equivalent in accordance with the intervals shown in table 3.3.1 above. The standard patrol shall comprise visual inspection of the following:

- any change of land use that may compromise conductor clearance, in accordance with NSP/004/011 "Overhead line conductor clearances";
- climbing aids around supports;
- encroachment into safety clearances from trees, buildings or other activities;
- all conductors, joints and terminations;
- all insulators;
- gross structural defects, e.g. bent members or poles significantly out of plumb; and
- danger of death signs.

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A risk classification shall be undertaken in accordance with the requirements in the policy for Risk Assessment of Overhead Lines (DSS/023).

3.3.4 Thermal Imaging

At every second inspection, a thermal imaging survey shall be carried out on both conductors and earth wires of all:

- terminations;
- dead ends;
- midspan joints; and
- bolted and bimetal connections.

This survey shall be carried out with the feeder at a minimum of 25% of full load, otherwise hot joints will not be discernible. It is recognised that this will require some switching, and perhaps some short-term security risk, but this is a lower impact than that of an undetected failure mode.

3.3.5 Stage 2 Condition Assessment

Lines shall be patrolled on foot or by equivalent means in accordance with the intervals shown in table 3.3.1 above. This shall include all the requirements of the stage 1 assessment, any high risk requirements appropriate to the site, and the following additional actions:

- concrete foundations shall receive a non-intrusive inspection for signs of cracking or water ingress. Unsound concrete shall be removed and remade. The muff and stub steelwork shall be painted with bitumastic paint if considered unsuitable to last until next inspection;
- EHV masts and towers with 'earth grillage' foundations with bare stubs buried in the ground or 'Malone anchor' type foundations shall be inspected for signs of corrosion. Steelwork shall be painted with bitumastic paint if considered unsuitable to last until next inspection;
- supports shall be assessed and remedial actions shall be carried out in accordance with NSP/004/112 "Guidance for the inspection and testing of wood and steel poles". Steel towers and masts shall be checked for steelwork / paint condition;

All new galvanised steel supports shall be considered for painting within 10 years of installation at a point when the steelwork surface can be suitably prepared and a protective paint system applied. Thereafter, steel supports shall be considered for re-painting when their condition is considered to be unsuitable to last until the next inspection. As a minimum, the need for re-painting shall be assessed at every 10 yearly inspection. All paint shall be supplied in accordance with NPS/001/021 "Technical specification for Paint systems for Overhead Line Towers";

- all insulators and fittings shall be visually assessed;
- stay insulators shall be visually assessed for integrity and inadvertent short-circuit (e.g. by barbed wire). Specifically, it shall be confirmed that all stays have insulators fitted and that they are positioned such that the insulator would be a minimum of 3m from the ground if the stay were to break, as detailed in NSP/004/104 "Guidance note on the types and installation requirements for stays". Where non-conformant, prompt replacement shall be arranged;
- arcing horn tips shall be visually assessed for burn back;
- surge arresters shall be visually assessed for damage/operation;

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- conductors, joints and terminations shall be checked for any obvious signs of damage, distress or deterioration;
- all safety signs, anti-climbing guards, jib height danger notices, designation plates or pole numbers shall be inspected and replaced if they are considered insufficient to last until the next inspection. All work shall be in accordance with NSP/004/109 "Guidance note on ACDs safety signs and labels required on overhead lines"; and
- a statutory risk assessment shall be carried out.

3.3.6 Helicopter Patrols (High Resolution Digital Imaging)

Helicopter patrols with digital imaging are particularly suited for the inspection of steel mast and tower overhead lines. These lines shall be patrolled by helicopter in accordance with the intervals shown in table 3.3.1 above. In circumstances where a helicopter patrol is not permissible (e.g. in a built up area) an equivalent dead or live top climbing inspection shall be carried out at the helicopter patrol interval.

The patrol shall comprise a visual inspection and acquisition of high resolution still digital images to record the condition of the following:

- anti climbing equipment – including barbed wire and barbed wire spacers;
- signing - including all statutory notices, safety signs, property plates, number plates and circuit identification plates;
- towers – including steelwork condition of tower legs, bracings, crossarms and tower peak. All step bolts shall also be examined for deterioration or obsolete design. Paint condition shall be recorded and an initial assessment made as to the feasibility of re-painting;
- earth wire and phase conductor fittings – including U-bolts, shackles, links, suspension and tension clamps, earth bonds. With some tower designs it may not be possible to determine the wear of suspension fittings and these tower types shall receive a dead top inspection in lieu of the helicopter inspection;
- insulators (including arcing horns) - signs of insulator deterioration due to age/wear, flashover or vandalism;
- earth wire and phase conductor – including bundled conductor spacers, mid span joints, repair sleeves and condition at conductor clamping positions; and
- downleads and cable sealing ends – including all safety signs and anti climbing equipment, decking, anti vandal cages, cable guards and downlead insulators.

3.3.7 Isolators

All EHV overhead line isolators shall be maintained as the equivalent ground-mounted apparatus, as defined in MNT/005.

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3.4 HV Overhead Lines

3.4.1 Summary of Intervals

Activity	Asset Type
	HV Overhead Lines
High Risk Site Inspection	2 Years
Stage 1 condition assessment	2 Years
Stage 2 condition assessment	10 Years

Table 3.4.1

Notes:

1. All condition and risk assessment data to be collected during inspections shall be entered onto an appropriate inspection sheet proforma or hand held device, then deprogrammed onto a database readily accessible across the business using standard asset identifiers
2. Inspections shall be combined as appropriate. For example, a high risk site shall receive both a stage 1 condition assessment and a high-risk site inspection every 24 months, and these shall be undertaken as a single visit wherever practicable. Similarly, every fifth inspection shall be a full condition assessment

3.4.2 High Risk Site Inspection

High risk sites shall be inspected in accordance with the intervals shown in table 3.4.1 above. During these inspections, particular attention shall be paid to:

- evidence of trespass or loitering;
- evidence of vandalism;
- classification of the risk associated with equipment;
- classification of the risks associated with the surrounding land;
- safety related assessment of equipment condition; and
- assessment of the existing mitigating measures at that site.

The detailed requirements for the high risk site inspection are described in the policy for Risk Assessment of Overhead Lines (DSS/023).

A high risk site where unauthorised access, trespass, theft or vandalism is evident will be classified as vulnerable. More frequent inspections of these sites shall be undertaken to monitor their status. The frequency of visits shall be determined depending on the particular risk that is evident. Information from RCM studies indicate that a quarterly inspection would be appropriate for these sites, although more frequent inspections may be required in particular circumstances.

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All sites with overhead line inadvertent contact warning signs shall receive an annual warning sign inspection as described in the policy for Risk Assessment of Overhead Lines (DSS/023). Such signs, and standard safety signs, shall be replaced if they are considered insufficient to last until the next inspection. All work shall be in accordance with NSP/004/109 "Guidance note on ACDs safety signs and labels required on overhead lines".

Consideration shall be given to removing the site from the vulnerable site register when no evidence of trespass activity is evident for more than two consecutive visits or additional control measures are employed.

3.4.3 Stage 1 Condition Assessment

Lines shall be patrolled by helicopter or equivalent in accordance with the intervals shown in table 3.4.1 above. The standard patrol shall comprise visual inspection of the following:

- any change of land use that may compromise conductor clearance, in accordance with NSP/004/011 "Overhead line conductor clearances";
- climbing aids around supports;
- encroachment into safety clearances from trees, buildings or other activities;
- all conductors, joints and terminations;
- all insulators;
- gross structural defects, e.g. bent members or poles significantly out of plumb; and
- danger of death signs.

A risk classification shall be undertaken in accordance with the requirements in the policy for Risk Assessment of Overhead Lines (DSS/023).

3.4.4 Stage 2 Condition Assessment

Lines shall be patrolled on foot or by equivalent means in accordance with the intervals shown in table 3.4.1 above. This shall include all the requirements of the stage 1 assessment, any high risk requirements appropriate to the site, and the additional following actions:

- visual inspection of all earths and electrical test of all safety-critical earths, i.e. those associated with:
 - o transformers;
 - o aerial switches and PMARCBs with operating mechanism/control cabinet below the anti-climbing guard; and
 - o cable terminations.

At pole-mounted substations, this shall include both HV and LV earths. Further, the location of HV & LV earths shall be checked. These connections must be separated by at least 90 degrees. On two-pole structures they shall be installed on separate legs of the pole. Where non-conformant, replacement shall be arranged;

- assessment of supports and remedial actions in accordance with NSP/004/112 "Guidance for the inspection and testing of wood and steel poles". Steel towers and masts (generally associated with de-rated lines) shall be checked for steelwork / paint condition;
- visual inspection of all insulators and fittings;

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- visual inspection of stay insulators for integrity and inadvertent short-circuit (e.g. by barbed wire). Specifically, it shall be confirmed that all stays have insulators fitted and that they are positioned such that the insulator would be a minimum of 3m from the ground if the stay were to break, as detailed in NSP/004/104 "Guidance note on the types and installation requirements for stays". Where non-conformant, prompt replacement shall be arranged;
- visual inspection of arcing horns and triggered spark gap arcing tips for burn back;
- visual inspection for damage/operation of surge arresters;
- visual inspection of conductors, joints and terminations for any obvious signs of damage, distress or deterioration;
- Fault Flow Indicator (FFI) replacement on a traveller basis;
- all safety signs, anti-climbing guards, jib height danger notices, designation plates or pole numbers shall be inspected and replaced if they are considered insufficient to last until the next inspection. All work shall be in accordance with NSP/004/109 "Guidance note on ACDs safety signs and labels required on overhead lines";
- visual inspection of pole mounted transformers for signs of damage and distress; and
- statutory risk assessment.

3.5 HV Pole-Mounted Plant

All aerial switches shall receive an energised condition assessment every 5 years. This shall include:

- mechanisms shall be exercised and correct operation confirmed;
- any encroaching vegetation shall be cleared;
- insulated links shall be visually inspected for integrity and inadvertent short-circuit (e.g. by barbed wire); and
- earth mat continuity and earth resistance shall be tested.

All aerial switches shall receive a scheduled overhaul every 10 years. This shall include all activities required at the 5-yearly condition assessment, plus:

- mechanism lubricant removed and re-applied if required;
- visual examination of contacts and replacement where unlikely to last until next maintenance; and
- visual examination of insulators and replacement where defective.

All oil filled pole mounted auto reclosers shall receive a scheduled maintenance every 4 years or 200 operations, whichever is sooner. This shall include:

- operational check;
- mechanism lubricant removed and re-applied if required;

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- oil tests for breakdown strength, moisture and acidity. If any of the concentrations defined in the table below are exceeded, the oil shall be changed:

H ₂ O		Acidity		Electrical Breakdown Strength (2.5mm gap)	
50	ppm	0.1	mgKOH/g	20	kV

- visual examination of contacts and replacement where unlikely to last until next inspection; and
- visual examination of insulators and replacement where defective.

All SF6 / vacuum pole mounted auto reclosers shall:

- where there is a cabinet below the anti-climbing guard, have earth mat continuity and earth resistance tested electrically every 5 years;
- receive a scheduled maintenance (including the above tests) according to battery type, specifically 5 years for Yuasa NP7-12 or equivalent and 10 years for EnerSys datasafe HX35 or equivalent. This shall include:
 - operational check (remote and local)[†];
 - visual inspection to confirm the integrity and security of:
 - recloser casing;
 - VT, specifically that where cast resin VTs have been installed that they have been fitted with insulator shrouds, that cover the electrical terminals but not the sheds of the insulators;
 - insulators;
 - antenna;
 - umbilical cables
 - making good of any defects; and
 - compulsory battery replacement with EnerSys datasafe HX35 or equivalent at each such maintenance visit. Asset records shall be updated to reflect the change of battery type.

[†] It is recognised that many reclosers operate frequently in the normal course of business. Any such operations in the preceding 12 months shall be taken to satisfy this requirement. However, some reclosers act only as remote switches, and operate only infrequently. This general policy requirement therefore remains

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3.6 LV Overhead Lines

3.6.1 Summary of Intervals

Activity	LV Open Wire Network	LV ABC Network	Eaves Mains	LV Above Ground Services
High Risk Site Inspection	2 Years	2 Years	2 Years	
Condition Assessment	10 Years	10 Years	10 Years	10 Years
Vegetation Inspection	2 years	4 years		4 years

Table 3.6.1

Notes:

1. All condition and risk assessment data to be collected during inspections shall be entered onto an appropriate inspection sheet proforma or hand held device, then deprogrammed onto a database readily accessible across the business using standard asset identifiers.
2. Inspections shall be combined as appropriate. For example, a high risk site shall receive both a stage 1 condition assessment and a high-risk site inspection every 24 months, and these shall be undertaken as a single visit wherever practicable. Similarly, every fifth inspection shall be a full condition assessment

3.6.2 High Risk Site Inspection

High risk sites shall be inspected in accordance with the intervals shown in table 3.5.1 above. During these inspections, particular attention shall be paid to:

- evidence of trespass or loitering;
- evidence of vandalism;
- classification of the risk associated with equipment;
- classification of the risks associated with the surrounding land;
- safety related assessment of equipment condition; and
- assessment of the existing mitigating measures at that site.

The detailed requirements for the high risk site inspection are described in the policy for Risk Assessment of Overhead Lines (DSS/023).

A high risk site where unauthorised access, trespass, theft or vandalism is evident will be classified as vulnerable. More frequent inspections of these sites shall be undertaken to monitor their status. The frequency of visits shall be determined depending on the particular risk that is evident. Information from RCM studies indicate that a quarterly inspection would be appropriate for these sites, although more frequent inspections may be required in particular circumstances.

All sites with overhead line inadvertent contact warning signs shall receive an annual warning sign inspection as described in the policy for Risk Assessment of Overhead Lines (DSS/023). Such signs, and standard safety signs, shall be replaced if they are considered insufficient to last until the next

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inspection. All work shall be in accordance with NSP/004/109 "Guidance note on ACDs safety signs and labels required on overhead lines".

A visual binocular check of pole top fittings shall be conducted. Fittings shall be programmed for replacement if considered unsuitable to last until next inspection.

Consideration shall be given to removing the site from the vulnerable site register when no evidence of trespass activity is evident for more than two consecutive visits or additional control measures are employed.

3.6.3 Stage 2 Condition Assessment

Lines shall be patrolled on foot or by equivalent means in accordance with the intervals shown in table 3.5.1 above. This shall include any high risk requirements appropriate to the site, and the additional following actions:

- visual check of supports for climbing aids and any tree encroachment;
- visual check of all conductors for encroachment into safety clearances from trees, buildings or other activities;
- assessment of supports and remedial actions in accordance with NSP/004/112 "Guidance for the inspection and testing of wood and steel poles". Steel towers and masts (generally associated with de-rated lines) shall be checked for steelwork / paint condition;

New galvanised steel poles need not be considered for painting until they reach their 30th anniversary. Thereafter, steel supports shall be considered for re-painting when their condition is considered to be unsuitable to last until the next inspection. As a minimum, the need for re-painting shall be assessed at every 10 yearly inspection. All paint shall be supplied in accordance with NPS/001/021 "Technical specification for Paint systems for Overhead Line Towers";

- Inspect for compound leaks from pole box and for cable gland armour corrosion.
- Check strategic or vulnerable poles with D iron and swing plates for corrosion.
- visual inspection of all insulators and fittings;
- visual inspection of stay insulators for integrity and inadvertent short-circuit (e.g. by barbed wire). Specifically, it shall be confirmed that all stays have insulators fitted and that they are positioned such that the insulator would be a minimum of 3m from the ground if the stay were to break, as detailed in NSP/004/104 "Guidance note on the types and installation requirements for stays". Where non-conformant, prompt replacement shall be arranged;
- visual inspection of conductors, joints and terminations for any obvious signs of damage, distress or deterioration;
- all safety signs, anti-climbing guards, jib height danger notices, designation plates or pole numbers shall be inspected and replaced if they are considered insufficient to last until the next inspection. All work shall be in accordance with NSP/004/109 "Guidance note on ACDs safety signs and labels required on overhead lines"; and
- statutory risk assessment.

3.6.4 Vegetation Inspection

A visual inspection shall be carried out every 2 years for open wire network and every 4 years for ABC network and above ground services. Vegetation issues shall be recorded and passed to Network services for rectification.

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Network Services shall implement and operate a vegetation management process that shall achieve and maintain clearances in accordance with NSP/004/011 "Overhead line conductor clearances" and DSS/024 "Management and Control of Vegetation Near Overhead Lines".

3.7 Assumptions

This policy is aimed at ensuring that overhead lines and pole mounted plant in operation on the network continues to operate in a safe and effective manner as a result of the preventative inspection and maintenance activities prescribed by this policy. Effective application of the required activities will contribute to minimising the number of asset failures experienced.

The following assumptions have been made but will be required to ensure the effective delivery of this policy:

- adequate budget provision will be made to secure delivery of these policy requirements;
- assets, in particular eaves mains, are identified and recorded so as to be able to programme inspections, record the outcome thereof, and schedule any remedial actions arising;
- the current business goals for the security, safety and continuity of operation of the network do not change;
- the detailed requirements for the inspection and maintenance activities that are identified within this policy are subsequently documented following the publication of this policy. In particular, battery types for reclosers are recorded and used to drive the generation of job cards at the appropriate frequency; and
- staff receive appropriate training.

3.8 Implementation and Monitoring

Several departments within CE Electric UK are accountable for the implementation and monitoring of the overhead line inspection and maintenance policy in accordance with the following table:

Designation	Responsibility
Network Construction Manager	Implementation of inspection and maintenance of overhead lines and pole mounted plant in line with the policy.
Safety Manager	Monitor compliance of those accountable for implementation of the overhead line inspection and maintenance policy.
Policy Production Manager	Responsible for monitoring compliance of all business divisions with this policy.

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3.9 Planned Policy Review

This policy shall be proposed for review on a biennial basis or at any time when external or internal influences drive a change in policy e.g. a change in legislation, learning points from the initial implementation stage, or as the optimum inspection and maintenance requirements are identified through the RCM (Reliability Centred Maintenance) process.

The following responsibilities shall apply to policy control and review:

Designation	Responsibility
Publication Manager	Responsible for issuing a quarterly report to the Policy Production Manager (or representative) detailing policies scheduled for biennial review within the next 6 months.
Policy Production Manager	Responsible for assessing the continued applicability of this company policy and for amending this document and communicating any changes in policy.

3.10 Superseded Documentation

This document supersedes the following documents, all copies of which shall be destroyed.

Ref	Title	Version
MNT/004	Policy for the Inspection and Maintenance of Overhead Lines	V2.0, Jun 2006

4 References

4.1 External Documentation

Reference	Title	Version and date
ESQC Regulations	The Electricity Supply, Quality and Continuity Regulations 2002.	No. 2665, 2002

4.2 Internal documentation

Reference	Title	Version and date
DSS/023	Policy for the Risk Assessment of Overhead Lines	V1.0, Oct 2003
DSS/024	Policy for the Management and Control of Vegetation Near LV Overhead Lines	V1.0, Nov 2003
MNT/005	Policy for the Inspection and Maintenance of Ground Mounted Plant	V1.0, Jul 2004
NPS/001/021	Technical specification for Paint systems for Overhead Line Towers	V1.0, Sep 2007
NSP/004/104	Guidance note on the types and installation requirements for stays	V2.0, Jan 2007
NSP/004/109	Guidance note on ACDs, safety signs and labels required on overhead lines	V1.0, Jul 2006
None	RCM Study – Overhead Line Summary of Tasks	

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5 Definitions

Term	Definition
LV	Low Voltage - a voltage up to and including 1000V
HV	High Voltage - a voltage greater than 1000V, but less than 33,000V
EHV	Extra High Voltage - a voltage at 33,000V up to and including 132kV
YEDL	Yorkshire Electricity Distribution plc
NEDL	Northern Electric Distribution Limited
CE Electric UK	CE Electric UK Funding Company (Encompassing NEDL/YEDL/IUS)
ESQCR	Electricity Safety, Quality and Continuity Regulations 2002
ABC	Aerial Bundled Conductor
FFI	Fault Flow Indicator
PNB	Protective Neutral Bond
RCM	Reliability Centred Maintenance

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6 Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
Dave Miller	Senior Asset Management Engineer	Dave Miller	30/07/08

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Andrew Ellam	Network Performance & Risk Manager	Andrew Ellam	20/08/08

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Mark Nicholson	Head of System Strategy	Mark Nicholson	30/07/08
Simon Dunn	Line Services Manager	Simon Dunn	11/08/08

6.4 CDS Assurance

I sign to confirm that this document has been assured for issue on to the CDS system.

		Sign	Date
Sean Johnson	Safety Engineer	Sean Johnson	04/09/08

6.5 Approval

Approval is given for the content of this document.

		Sign	Date
Peter McCormick	Director of Group Health, Safety and Environment	Peter McCormick	22/08/08

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6.6 Authorisation

Authorisation is granted for publication of this document.

		Sign	Date
Mark Drye	Director of Asset Management	Mark Drye	31/08/08